

planes closer means that signal traces have to be located on the outermost planes, which can cause increased emission and susceptibility. Fig. 13 shows a traditional stacking in a four-layer board where layers are spaced equally. Ground and power planes (layers 2 and 3) are sandwiched between signal layers (layers 1 and 4).

This type of distribution is called microstrip distribution and provides excellent decoupling capacitance between power and ground planes since these are located close together. This, along with reduced inductance, provides a 20dB to 30dB improvement for emission and susceptibility over the two-layer board. However, since traces are on the outside of the board, there exists potential emission and susceptibility.

To reduce potential emission and susceptibility problems of microstrip distribution, power and ground planes can be placed on the outside, while signal traces are sandwiched between the two. Such type of distribution is called stripline distribution. As shown in Fig. 14, signal traces on layers 2 and 3 are very close and are oriented perpendicular to each other (for reducing cross-talk). However, in this type of stacking, power and ground planes are so far apart and their distributed capacitance so low that there is negligible decoupling. This has to be compensated by providing larger decoupling capacitors.

Another drawback of stripline distribution is that of increasing difficulty as far as hand repairs and rework is concerned, especially during prototype development.

Board stacking of six-layer PCB.

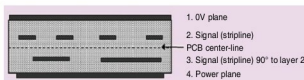


Fig. 14: Stripline configuration

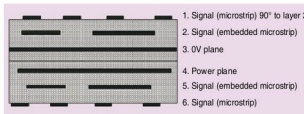


Fig. 15: Board stacking of a six-layer PCB

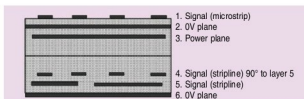


Fig. 16: Another type of board stacking of a six-layer PCB

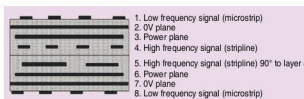


Fig. 17: Board stacking of an eight-layer PCB

A six-layer board provides more design options to improve EMC characteristics if stacking is done judiciously. Fig. 15 shows board stacking for a six-layer board. Here, power and 0V planes are located at the centre on either side of PCB centre line, with two signal-track layers above and below these. This type of stacking does not provide much EMC performance improvement over a four-layer board, except that there are just two additional signal layers to play with.

Some improvement in performance can be gained by using the type of stacking shown in Fig. 16, where positions of inner signal and power plane layers have been interchanged. This causes lowering of power supply decoupling, but can provide good shielding

to high-speed signals if these are routed on the inner layers (layers 3 and 4). Again, tracks on these two inner layers have to be laid perpendicular to avoid cross-talk.

Board stacking of eight-layer PCB. Symmetrical stackup using an eight-layer PCB can meet all EMC requirements, without having to resort to compromises and additional efforts taken for unsymmetrical stackup. A traditional eight-layer symmetrical stackup is shown in Fig. 17, which consists of four signal layers and two pairs of 0V power plane. Since in both power plane pairs, 0V and power planes are located close together, excellent decoupling capacitance is obtained. Very good shielding can be provided by routing the high-speed, high-frequency tracks on innermost layers 4 and 5. Since the tracks on any layer have a plane nearby, image plane effect so created leads to low

decoupling inductance. Overall, an eight-layer board provides the most effective high-speed performance with 20dB or more improvement over a six-layer board.

Conclusion

So far we have discussed the various aspects of a good PCB design to take care of both signal integrity and EMI issues. A good PCB design is always the starting point of a good EMC design and is the most important too, since it reduces performance requirements of filters and shields, thus contributing a long way in reducing the product cost and keep the product competitive.

This article is an extract from a book by the author. The next part will cover grounding.